



London Ambulance Service



NHS Trust

Support the management of trauma in the emergency and urgent care setting Component 8

Trauma Overview



Trauma - Component 8

Session Outline

- Major Trauma – the risk of serious injury
- DR C_ABCDE Approach
- Supporting the EAC/paramedic & keeping things moving!
- Designed to complement existing knowledge on the management of simple wounds and minor injuries
- Transfer existing skills into the context of major trauma



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Key messages before we start!

- Scene safety
- Minimal on scene times!
- Pre-empt what paramedic is going to do!
- Have kit ready
- Expedite packaging and removal
- Prevent paramedic from performing unnecessary interventions and assist them where needed
- Tell paramedic about any of your findings

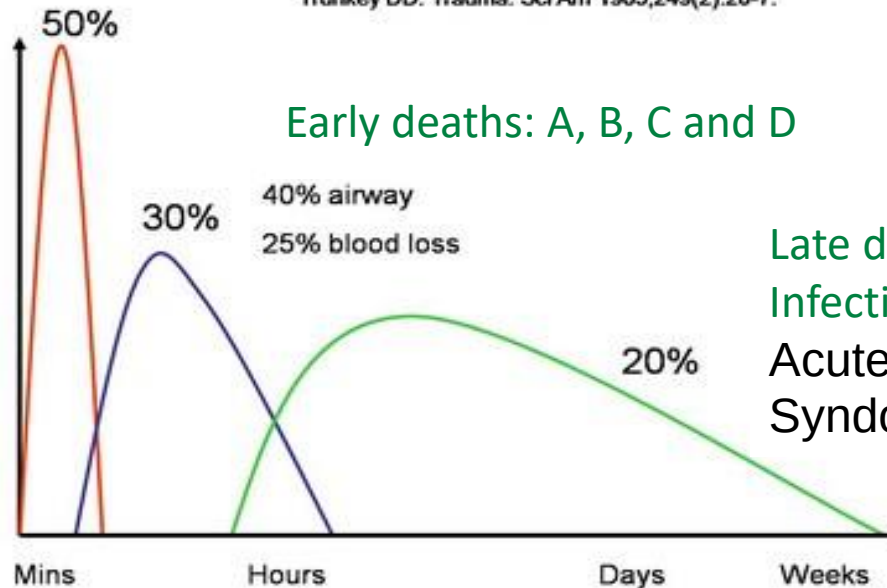
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Trauma Deaths

Immediate fatal injury

Trimodal Distribution of Death

Trunkey DD. Trauma. Sci Am 1983;249(2):20-7.



Early deaths: A, B, C and D

Late deaths

Infection, ARDS

Acute Respiratory Distress Syndrome

What can be done to reduce deaths in each phase?

**The Major Cause of
Death in Trauma is
Blood Loss**



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Compensation for Blood Loss

- Most trauma victims are young and healthy
- They compensate very well for blood loss
 - Vasoconstriction
 - Increased heart rate
 - Increased respiratory rate
- Vital signs can be misleading

In young, healthy adults
these changes are often subtle and
may only be seen late!



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Changes in Vital Signs

- “Text-Book” normal heart rate:
60 – 80 / minute
- A heart rate of 100 is only slightly outside this range...
yet may be almost double the patient’s resting rate



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Changes in Vital Signs

- “Text-Book” normal breathing rate
12 - 20 / minute
- A breathing rate of 24 is only slightly outside this range... yet may be double the patient’s resting rate

A young persons ability to compensate for
major trauma

+

The fact that we do not always witness and
appreciate the forces involved

=

Serious injuries may be missed



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Who is at risk of serious injury?

- Young, healthy adults
 - Alcohol & Drugs
- Older people
 - Less resilient
 - Multiple medical problems and medications
- Children
 - No concept of danger

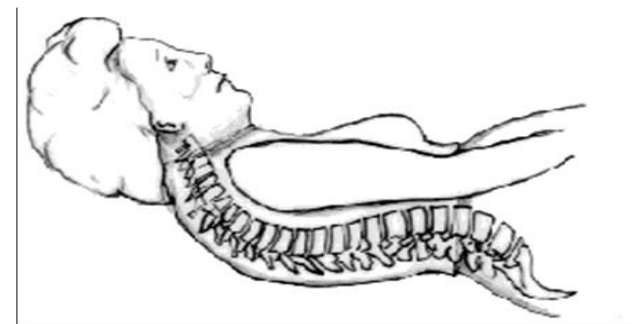
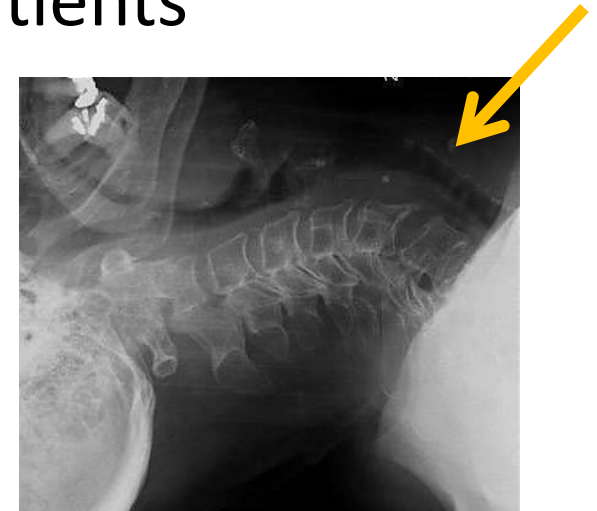


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- Affects older patients
- Increased rate of significant injuries from falls from standing
- Elderly and fall downstairs is bad
- Anti-coagulant medication can complicate the injury

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- Major trauma following low-mechanism injury is common in older patients
 - Careful neurological assessment
 - Immobilisation in a **neutral** position for the patient
 - Not applying a collar is not 'clearing the spine'



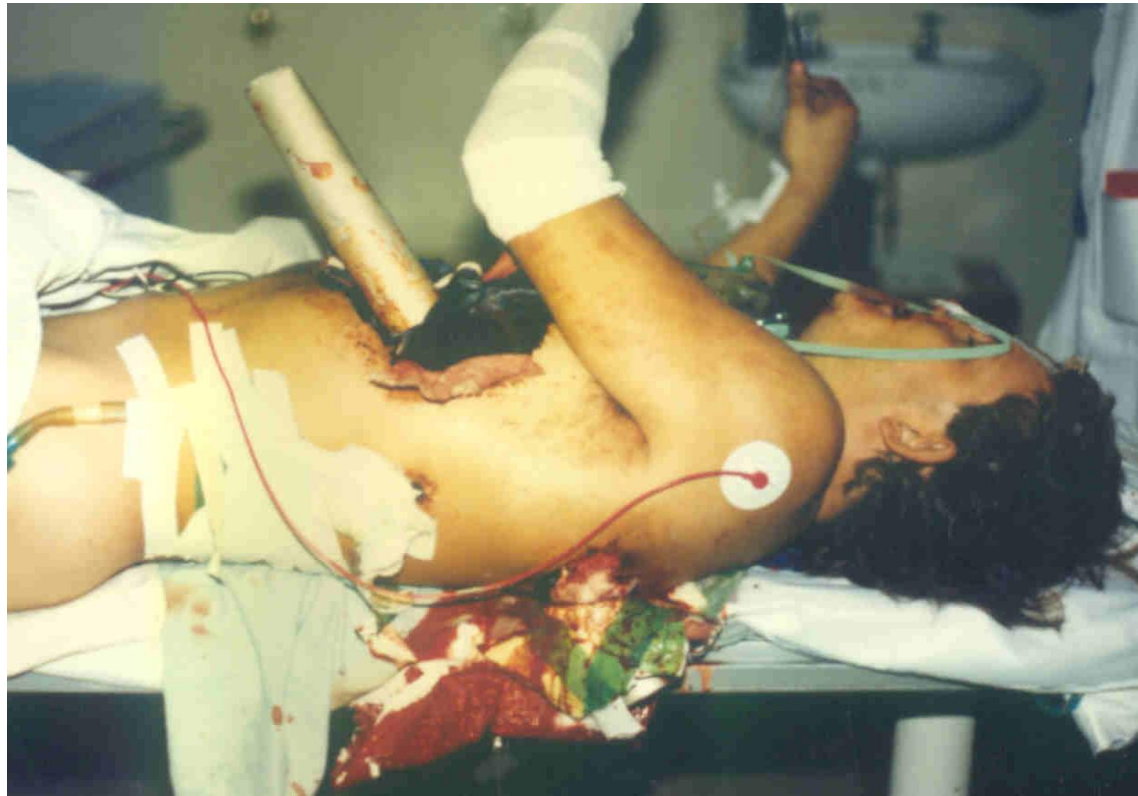


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Why do we miss seriously injured patients?

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Sometimes it is obvious





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But...

- Time to scene ?
- Experience ?
- Mechanism ?
- Compensation ?

**GO
NOW!**

Minimise on-scene time for Major Trauma

“minutes” for penetrating trauma

≤ 20 minutes for blunt trauma



Secondary survey - Component 3

Mechanism of Injury

Understanding how traumatic injury occurs

Penetrating injury
cannot be fixed on scene
or in the ambulance

Keep on-scene time to the absolute
minimum

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... when you can't go now





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Get Everything Ready

- Kit out
- Trolley out
- Ambulance pointing in the right direction
- Heating on
- Exit route planned and clear
- Family and police briefed



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Scene Management

- Involve everyone
- Match the skill level to the task
- Who is holding the head?
- Have an overview
- Anticipate & multi-task
- Communicate



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**Good Trauma Care
is a Team Sport**







Trauma - Component 8

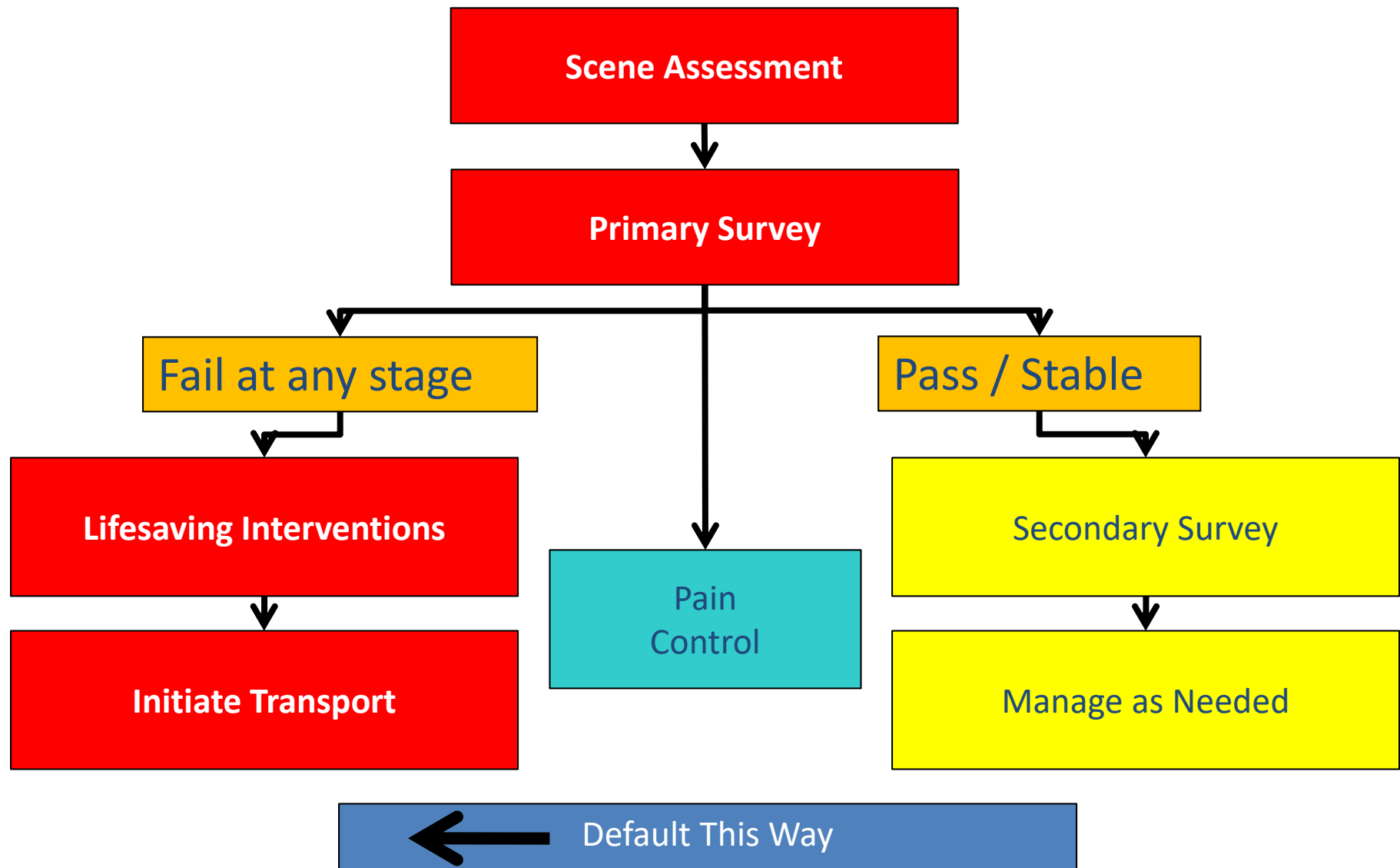
The approach to the seriously injured patient

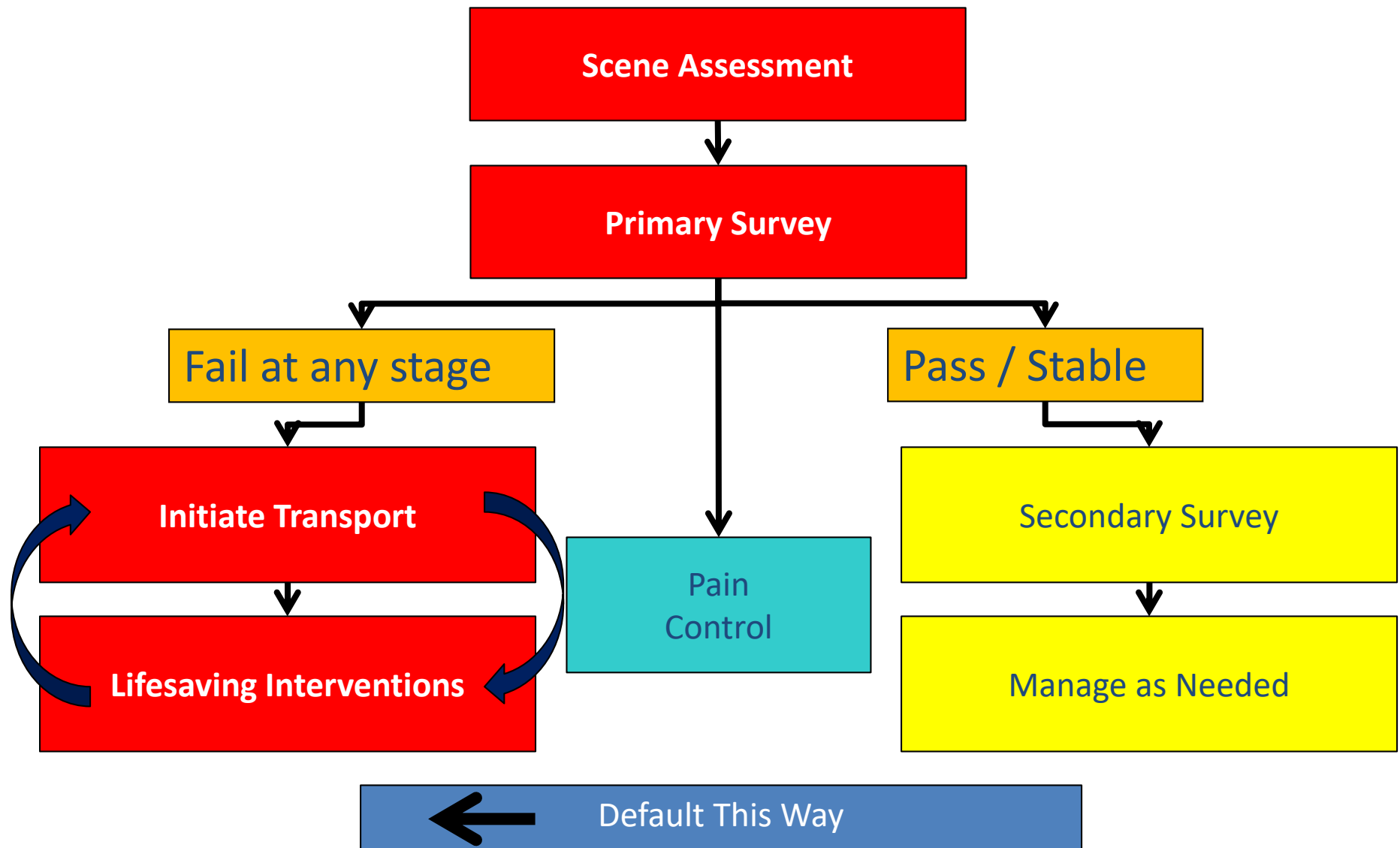


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The first 10 minutes are vital

- Consistent approach
- Rapid identification of life-threatening injuries
- Assessment and management are simultaneous
- Assume the worst and exclude or treat it







Trauma - Component 8

The Primary Survey



D – Danger

R – Response

C - Catastrophic haemorrhage

A – Airway

C – C – spine consideration

B – Breathing

C – Circulation (Blood on the Floor PLUS Four More)

D – Disability

E – Expose and evaluate

All seriously injured patients
must get 100% Oxygen
as soon as their airway
is clear

Any patient with a
serious head injury
or
loss of consciousness
has a spinal injury
until proven otherwise



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Your role: In addition....

- Anticipate
- Monitor
- Prepare
- Plan

Keep things moving !



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Trauma Priority Call: PD 09

- C** • Call sign
- A** • Age
- T** • Time of injury
- M** • Mechanism of injury
- I** • Injuries found and suspected
- S** • Signs (vital)
- T** • Treatment given or required



Trauma - Component 8

Key Messages

- Be thorough and effective with airway management
- Don't get distracted
- Expose the patient – if you don't look you won't find (keep them warm afterwards)
- Hypoxia and/or hypotension will kill the brain injured patient
- The first clot is the best one
- Minimal movement of the seriously injured (don't let this slow you down though!)



Trauma - Component 8

Key Messages

- Get the patient to the
 - right hospital
 - in the right time
 - first time
- If you are tasked to significant trauma – think about your options for support early
- Pre-empt what the paramedic will do and ensure kit ready and assist where possible
- Tell the paramedic of your findings

There are not many things
that can be fixed on scene
or in the ambulance

Keep on-scene time to the absolute
minimum



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Questions?



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Summary

- Good trauma care requires:
 - **Rapid assessment**
 - **Immediate interventions**
 - Airway, Breathing, Circulation
 - **Attention to detail**
 - **Continual reassessment**
 - **Teamwork**